

JEDDAH KING ABDULAZIZ INTL, Saudi Arabia

WMO: 410240

Lat: 21.680N

Lon: 39.157E

Elev: 48

StdP: 14.67

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 61.1	(c) 63.0	(d) 33.9	(e) 28.6	(f) 79.9	(g) 39.5	(h) 35.7	(i) 80.5	(j) 25.1	(k) 78.3	(l) 22.5	(m) 77.5	(n) 5.9	(o) 30	(p) 0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 16.2	(c) 105.8	(d) 74.7	(e) 103.8	(f) 75.6	(g) 102.0	(h) 76.2	(i) 85.7	(j) 95.0	(k) 84.4	(l) 93.8	(m) 83.4	(n) 93.1	(o) 13.4	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.9	177.6	92.3	82.3	168.5	90.9	80.8	160.2	90.2	50.5	95.5	48.8	94.7	47.7	92.5	95.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 21.4	(b) 19.0	(c) 17.1	(e) 58.0	(f) 114.1	(g) 2.2	(h) 3.9	(i) 56.4	(j) 116.9	(k) 55.0	(l) 119.1	(m) 53.8	(n) 121.3	(o) 52.2	(p) 124.1		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			48.7	88.4	2.0	2.9	47.3	90.5	46.1	92.2	45.0	93.9	43.5	96.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	84.4	75.0	76.2	78.6	83.4	87.8	89.5	92.1	92.4	90.0	87.1	82.3	78.0
	DBStd	6.89	3.76	4.01	4.22	4.32	3.93	3.21	2.40	2.32	2.38	2.43	3.04	3.23
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12566	776	734	887	1003	1172	1186	1307	1315	1199	1150	969	868
	CDD65	7091	311	314	422	553	707	736	842	850	749	685	519	403
	CDH74	93049	2209	2500	4074	6786	10028	11073	13415	13416	11117	9169	5820	3442
(13)	CDH80	53084	592	800	1628	3482	5990	6956	8999	8967	6865	5071	2602	1132
(14)	Wind	WSAvg	8.1	8.4	8.6	9.0	8.5	8.9	8.3	8.4	7.8	6.6	7.1	7.7
(15) Precipitation	PrecAvg	2.4	0.5	0.2	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.1	0.9	0.4
	PrecMax	11.4	2.8	1.6	0.9	1.9	0.2	0.0	0.3	0.7	0.2	1.1	9.4	2.5
	PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.2	0.8	0.4	0.2	0.4	0.0	0.0	0.1	0.1	0.0	0.2	1.9	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.0	93.1	98.3	102.3	107.8	110.8	107.7	106.1	104.4	105.3	97.1	91.8
		MCWB	69.1	68.3	70.8	72.5	74.6	75.1	74.2	76.0	77.4	72.4	71.1	69.8
	2%	DB	87.9	89.8	93.5	98.6	102.6	104.1	105.5	103.8	100.7	100.0	94.7	89.8
		MCWB	70.6	69.7	71.0	73.3	75.1	74.8	74.7	77.7	78.8	76.0	73.3	70.6
	5%	DB	85.9	87.6	90.7	95.4	100.1	101.6	103.6	101.9	98.8	97.0	92.8	87.9
		MCWB	69.7	70.5	71.0	74.0	75.6	75.7	75.1	78.3	79.9	77.5	73.6	70.8
	10%	DB	83.7	85.1	87.9	93.2	97.1	98.8	101.6	100.0	97.0	95.1	90.6	86.1
		MCWB	69.1	69.9	70.8	73.5	75.3	76.6	75.9	79.3	80.5	77.8	73.7	70.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.8	77.3	78.4	80.9	84.3	84.2	85.7	88.0	87.0	84.5	80.6	77.9
		MCDB	84.5	86.2	88.7	92.7	97.6	97.2	96.4	96.9	96.5	93.4	90.0	85.9
	2%	WB	74.7	75.3	75.9	78.9	81.2	82.4	83.9	86.2	84.7	82.5	78.9	76.0
		MCDB	82.7	84.2	86.6	90.2	93.7	94.3	93.8	94.8	94.0	91.6	88.0	84.6
	5%	WB	73.2	73.8	74.5	77.3	79.8	81.1	82.5	84.7	83.9	81.4	77.4	74.8
		MCDB	81.2	82.3	85.2	88.9	92.2	92.7	92.9	93.6	93.1	90.5	86.6	83.4
	10%	WB	71.8	72.5	73.2	75.9	78.5	80.2	81.3	83.7	82.8	80.5	76.5	73.3
		MCDB	80.2	81.0	83.7	87.9	91.0	91.9	92.5	93.1	91.8	89.6	85.7	81.8
(35) Mean Daily Temperature Range	5% DB	MDBR	16.6	17.7	19.0	19.7	20.6	20.8	19.9	16.2	16.9	19.3	17.5	16.6
		MCDBR	19.4	20.4	22.5	22.3	24.1	24.7	23.2	19.5	19.2	22.0	19.9	18.9
		MCWBR	10.1	10.4	10.1	10.1	10.7	10.6	10.8	9.7	8.7	9.4	8.8	9.8
	5% WB	MCDBR	16.6	17.5	18.7	19.4	21.0	19.8	18.7	15.4	16.6	19.6	17.9	16.7
		MCWBR	9.5	9.7	10.1	10.4	10.7	10.0	10.4	9.0	8.3	9.0	8.3	9.0
(40) Clear-Sky Solar Irradiance	taub		0.471	0.517	0.533	0.555	0.596	0.616	0.649	0.670	0.630	0.569	0.504	0.481
	taud		2.085	1.951	1.902	1.877	1.772	1.727	1.700	1.666	1.723	1.852	2.014	2.060
	Ebn at Noon		250	246	249	245	233	226	220	215	221	228	238	241
	Edh at Noon		48	57	63	65	72	75	77	80	74	62	50	47
(44) All-Sky Solar Radiation	RadAvg		1424	1711	2055	2259	2376	2442	2329	2191	2053	1828	1519	1359
	RadStd		76	75	68	80	71	49	61	52	60	49	63	50

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.06	+2.03	-3.24	N/A	-0.65	-0.49	N/A	N/A	+362
(47) Regional (2 neighbors)		+0.86	+1.26	-2.23	N/A	N/A	N/A	N/A	N/A	+310

Nomenclature: See separate page